Sosna Worku Achamyeleh

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RESEARCH SUMMARY

Research on whether deep learning models' confidence and explanations can be trusted when the data shifts. Current work spans chest X-ray classification under cross-hospital shift, comparing vision-pretrained and EEG-pretrained models for decoding EEG from unseen users, representation analysis linking deep networks to human visual cortex, and analytical estimation of model performance before inference.

Research Interests: Trustworthy AI, Distribution Shift and Robustness, Uncertainty Quantification and Calibration, Explainability, Foundation Models, Representation Analysis, Multimodal Learning

EDUCATION

The George Washington University — Washington, DC, USA
M.S. in Artificial Intelligence & Machine Learning — Aug 2025 – May 2027

- **GPA:** 3.78/4.00
- **Black in Robotics + 3M Tuition Fellowship ($26,115):** Awarded to one of the top two applicants.
- **Master's Thesis:** Whether domain-specific pretraining makes EEG decoding more trustworthy for unseen users (in progress), advised by Professor Xiaodong Qu.
- **Coursework:** Machine Learning, Neural Networks & Deep Learning, Trustworthy AI, Design & Analysis of Algorithms, Introduction to Big Data & Analytics, Cloud Computing, Computer System Architecture, Advanced Software Paradigms

Addis Ababa University — Addis Ababa, Ethiopia
B.S. in Software Engineering and Computing Technology (Software Engineering Stream) — Oct 2019 – Jul 2024

- **GPA:** 3.62/4.00 (Great Distinction)
- **Coursework:** Natural Language Processing, Fundamentals of Artificial Intelligence, Machine Learning and Big Data, Data Structures & Algorithms

PUBLICATIONS AND MANUSCRIPTS

- **Achamyeleh, S. W.** Reliability and Explainability of Vision Transformers Under Medical Distribution Shift: A Comparative Study on Chest X-Ray Classification. *International Conference on Medical Imaging and Computer-Aided Diagnosis (MICAD 2026)*, Edinburgh, UK, October 2026. **Oral presentation.**
- **Achamyeleh, S. W.**, Ma, R., and Arora, A. Estimating Viterbi Decoding Accuracy Using Emission and Transition Similarity. Submitted to the *19th International Conference on Agents and Artificial Intelligence (ICAART 2027)*.
- **Achamyeleh, S. W.** SubTrust: Does Domain-Specific Pretraining Make EEG Decoding More Trustworthy for Unseen Users? Paper proposal submitted to the *International Conference on Human-Computer Interaction (HCII 2027)*, AI in HCI track.
- **Achamyeleh, S. W.**, Saralebettu, P. P., and Prabhakara, S. M. Transformers Are More Accurate but More Confidently Wrong in Stress Detection on Reddit. **Manuscript**, 2026.

RESEARCH PROJECTS

Reliability and Explainability of Vision Transformers Under Medical Distribution Shift — Jun 2026 – Sep 2026
The George Washington University, independent research, sole author (MICAD 2026) [Code]

- Asked whether chest X-ray models that perform well on familiar data still hold up on a new hospital's data, comparing ViT-B/16 and ResNet-50 on predictive performance, calibration, uncertainty, and explainability.
- Built a cross-institutional test by manually mapping seven diagnostic conditions shared by NIH ChestX-ray14 and CheXpert, whose label sets differ, then trained both models in PyTorch on NIH data and evaluated them out of distribution on CheXpert.
- ViT reached higher in-distribution mean AUC (0.800 vs 0.745), yet both models degraded by a similar amount under shift, with ViT far more vulnerable on Pneumothorax (0.227 vs 0.108 AUC drop). Expected calibration error rose from 0.006 to 0.040 for ViT-B/16 and from 0.003 to 0.035 for ResNet-50, and MC-Dropout did not give a usable out-of-distribution uncertainty signal.
- Found that ViT attention rollout and ResNet-50 Grad-CAM produced very different localization patterns (diffuse vs more localized), showing that predictive performance alone is insufficient to characterize reliability under distribution shift.

Estimating Viterbi Decoding Accuracy Using Emission and Transition Similarity — Feb 2026 – Sep 2026
The George Washington University, supervised by Professor Amrinder Arora, first author (submitted to ICAART 2027)

- Developed a parameter-based estimator of Viterbi decoding accuracy for hidden Markov models (HMMs), extending the emission-similarity estimator of Zhao et al. (2021) with transition similarity, so accuracy can be predicted without running the algorithm.

- Reasoned that states that transition alike should be as hard to tell apart as states that emit alike, and combined cosine similarity between emission rows and between transition rows, weighted by the stationary distribution.
- Evaluated the estimator on 189 systematically constructed HMMs spanning 3 to 8 hidden states and 4 to 14 observation symbols, reducing mean absolute error from 0.0378 to 0.0272, about 28% below the emission-only baseline.
- Implemented and tested the estimators, ran the experiments, analyzed the results, and wrote the first draft of the paper.

SubTrust: Does Domain-Specific Pretraining Make EEG Decoding More Trustworthy for Unseen Users? Sep 2026 – Present
The George Washington University, Master's thesis advised by Professor Xiaodong Qu (proposal submitted to HCII 2027) [Code]

- Proposed SubTrust, a framework testing whether EEG decoders stay trustworthy, not just accurate, for new users.
- Designed a comparison of ImageNet-pretrained ViT-B/16 on EEG time-frequency images, LaBraM on raw EEG, and EEGNet trained from scratch, with randomly initialized controls to investigate the contribution of pretraining.
- Will evaluate on unseen subjects (31-subject motor imagery dataset, five-fold subject-level cross-validation) and on a 51-subject dataset from another site, measuring accuracy, calibration, selective prediction, and whether temperature scaling transfers.
- Proposed the Saliency Plausibility Score, the share of integrated-gradient saliency in mu and beta bands over the contralateral motor cortex, compared against random-region and randomized-weight baselines.
- Completed ViT-B/16 feature extraction for all 31 subjects; a five-subject check confirmed the expected contralateral desynchronization.

Investigating Face-Selective Visual Information Using Deep Neural Network Ablation Jul 2026
Neuromatch Academy, Computational Neuroscience, 6-person team

- Investigated whether face-selective units in deep networks carry the information that predicts fMRI responses in the fusiform face area (FFA), the brain's face-selective region.
- Built the project's voxel-wise ridge encoding models across nine visual regions from AlexNet and VGGFace2-trained ResNet-50 layer activations (Algonauts 2021, 1,000 video clips), with shuffled-label, random-seed, and per-voxel RidgeCV checks.
- Showed that AlexNet's deepest layers predicted FFA better than any ResNet-50 layer (0.266 vs 0.184), and that ResNet-50 accuracy dropped with depth, the opposite of AlexNet's pattern.
- The team's ablation of the 50 most face-selective AlexNet units degraded FFA prediction significantly more than random or activity-matched controls ($p < 10^{-4}$).

Transformers Are More Accurate but More Confidently Wrong in Stress Detection on Reddit Jan 2026 – May 2026
The George Washington University, 4-person team, first author on manuscript

- Compared classical machine learning and transformer models for detecting stress in Reddit posts (Dreaddit dataset).
- Contributed to building Naive Bayes, TF-IDF Logistic Regression, and DistilBERT models.
- Designed the evaluation to be statistically defensible, using six stratified train/validation/test splits with McNemar's test, the Wilcoxon signed-rank test, and the Nadeau–Bengio corrected t-test.
- Showed that DistilBERT performed best at 0.793 macro F1 and 0.880 ROC-AUC, outperforming the classical baselines.

Threat Detection in Spoken Amharic Language Sep 2023 – Jun 2024
Addis Ababa University, undergraduate capstone supervised by Dr. Beakal Gizachew, 5-person team

- Contributed to building a real-time speech classification system distinguishing threatening from non-threatening spoken Amharic, a language with almost no existing speech resources.
- Collected and annotated 5,800 audio recordings, augmented with time stretching, pitch shifting, noise, and time shifting.
- Extracted MFCC, Chroma, spectral, and zero-crossing features, then evaluated CNN, RNN, and LSTM architectures.
- Showed that a CNN performed best, reaching 91% test accuracy and 0.89 F1, demonstrating that real-time threat detection in spoken Amharic is feasible.

TEACHING EXPERIENCE

Addis Ababa University Addis Ababa, Ethiopia
Assistant Lecturer (Part-time, Computer Programming) Mar 2025 – Aug 2025

- Taught an undergraduate Computer Programming course through lectures and labs, covering Python, loops, functions, object-oriented programming, and algorithms.
- Followed each class with a 20 to 30 minute live coding session, writing and debugging that day's material in front of students so they could see the concepts work in practice.
- Guided students from flowcharts to working Python programs and supervised their group projects.

AddisCoder Addis Ababa, Ethiopia
Teaching Assistant (Data Structures and Algorithms) Jul 2024 – Aug 2024

- Supported 99 high school students from across Ethiopia in AddisCoder, a programming and algorithms program founded by Professor Jelani Nelson (UC Berkeley).

- Helped students implement algorithms and data structures, including dynamic programming, in lab sessions, and delivered mini-lectures.

Ethioware EdTech Initiative Remote
Back-end Development Instructor (Volunteer) Feb 2024 – Mar 2024

- Trained 54 women across multiple African countries in backend development with C#, ASP.NET Core, PostgreSQL, and GitHub through project-based instruction, including a mini-project in which they built a task management system backend.

PROFESSIONAL EXPERIENCE

Phoenixopia Solutions PLC Addis Ababa, Ethiopia
Software Developer Mar 2025 – Jul 2025

- Developed responsive Next.js and Tailwind CSS web applications, using routing and image optimization to improve performance.
- Integrated backend REST APIs to retrieve and display dynamic data.

Omdena San Jose Chapter Remote (San Jose, CA, USA)
Task Lead, Junior Machine Learning Engineer (Volunteer) Oct 2024 – Mar 2025

- Led a team of 25 junior machine learning engineers on a predictive modeling project estimating risk of Coronary Artery Disease and Heart Failure from National Health and Nutrition Examination Survey (NHANES 2017–2020) data.
- Guided the development of preprocessing workflows, handling missing data with Multivariate Imputation by Chained Equations (MICE) and selecting features with Recursive Feature Elimination (RFE).
- Supervised exploratory data analysis of relationships between health indicators and disease outcomes in Matplotlib and Seaborn.
- Guided the development and evaluation of Random Forest, XGBoost, and Logistic Regression models; XGBoost performed best at 88% accuracy, 0.86 precision, 0.84 recall, and 0.85 F1.
- Mentored team members on preprocessing, machine learning workflows, and model evaluation.

TECHNICAL SKILLS

Programming: Python, C#, JavaScript, SQL
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
Data & Analysis: NumPy, Pandas, Matplotlib, Seaborn
Research Methods: Calibration and uncertainty evaluation (ECE, MC-Dropout), out-of-distribution evaluation, statistical testing (McNemar, Wilcoxon, corrected t-test), interpretability (Grad-CAM, attention rollout), voxel-wise encoding models
Tools: Git, GitHub, Jupyter, Google Colab

LEADERSHIP EXPERIENCE

Africa to Silicon Valley (A2SV) Addis Ababa, Ethiopia
Head of Communications Education Nov 2023 – Mar 2025

- Led the Communications Education program, designing and delivering training in professional communication, presentation, and technical interview skills.
- Mentored and taught 205 university students from more than 20 African countries through interactive lectures and workshops.

AWARDS AND ACHIEVEMENTS

- Best Poster Presentation (Team Award), AAU Research Week 2024, for *Threat Detection in Spoken Amharic Language*.
- Top 2% Finalist, 2023 Aspire Leaders Program, selected from a global applicant pool.
- Third-place team award ($1,250) for impactful real-world projects, A2SV internship program with Eskalate LLC.
- Selected among the top 40 students in Ethiopia to represent the country in Huawei's Seeds for the Future program, 2021.

POSTERS AND PRESENTATIONS

- *Reliability and Explainability of Vision Transformers Under Medical Distribution Shift*, MICAD 2026 (hybrid conference, Edinburgh, UK), oral presentation delivered online, October 22, 2026.
- *Benchmarking Classical and Transformer-Based NLP Models for Stress Detection*, GWU SEAS R&D Showcase, April 2026.
- *Threat Detection in Spoken Amharic Language*, Addis Ababa University Research Week, May 2024.